

Debabrata Sarkar

Technology Analyst, Infosys (Kolkata)

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PROFILE SUMMARY

Technology Analyst with **4+ years** of experience building scalable, **AI-powered solutions** at Infosys. Core expertise includes **Generative AI**, Python, LangChain, **Prompt Engineering**, and robust back-end API development using Flask, primarily deployed on **Google Cloud Platform (GCP)**. Proven success in designing and implementing Retrieval-Augmented Generation (**RAG**) pipelines, integrating **Gemini Flash LLM** to deliver accurate, context-aware enterprise responses, and optimizing model performance through LLM fine-tuning. Passionate about solving real-world business problems using AI to drive innovation and enhance customer experience.

CORE SKILLS

- **Programming:** Python, SQL
- **AI/LLMs:** LangChain, OpenAI, Gemini Flash, Hugging Face, LLM Fine-tuning, RAG
- **Expertise:** Prompt Engineering, Semantic Search, Conversational AI (**Dialogflow**)
- **Cloud/DevOps:** **Google Cloud Platform** (Vertex AI, Cloud Build, Workbench, Storage, IAM)
- **Databases:** PostgreSQL, Qdrant (Vector DB)
- **Other:** Flask, REST APIs, UiPath, Streamlit

PROFESSIONAL EXPERIENCE

Technology Analyst, Infosys

Oct 2024 – Present

- **Developed and deployed a robust Semantic Search Application** using **Python** and **LangChain** to process proprietary data, achieving a measurable **20% reduction** in time-to-insight for internal users.
- **Architected and productionized RAG pipelines**, integrating **Gemini Flash LLM** and Qdrant Vector DB to generate highly accurate, context-aware responses, cutting reliance on manual lookups.
- **Led Prompt Engineering initiatives** (few-shot, Chain-of-Thought) and **LLM fine-tuning** efforts that optimized model performance, raising accuracy by **10%** for the AI-driven Contract Compliance Tool.
- **Built and secured high-throughput backend services** using **Flask** and REST APIs, handling critical data communication and supporting scalable front-end interfaces.
- **Managed MLOps workflows** leveraging **GCP** services: **Vertex AI Workbench** for scripting, **Cloud Build** for automated CI/CD, and GCP Storage for data governance.
- **Integrated Conversational AI framework Dialogflow** to structure and deploy virtual agent components, including configuring Intents, Webhooks, and Fulfillment logic for end-to-end user interaction flows.

Senior System Engineer, Infosys

Oct 2023 – Sep 2024

- **Automated critical financial and administrative enterprise workflows** using the **UiPath** RPA platform, encompassing complex operations like OCR, form filling, and email/SMS communication.
- **Delivered efficiency improvements** by reducing manual back-office effort by an estimated **20%** through the design and deployment of robust **web scraping solutions** using Python for automated data acquisition.
- **Managed and optimized SQL database structures** for high-volume data processing tasks, improving query speed and ensuring data integrity across multiple legacy systems.

System Engineer, Infosys

Oct 2021 – Sep 2023

- **Provided continuous support for high-availability enterprise production systems** hosted on **Google Cloud Platform**, focusing on maintaining system reliability, security, and compliance standards (IAM, Kubernetes).
- **Executed proactive incident management** by monitoring application workloads and performing root cause analysis using **GCP log-explorer** for critical production applications, minimizing downtime and service interruptions.
- Contributed to CI/CD processes using **Cloud Build** for continuous deployment of application updates, ensuring rapid and safe delivery of new features and patches.

EDUCATION

Meghnad Saha Institute of Technology, MAKAUT University

08/2017 – 08/2021

Bachelor of Technology in Civil Engineering | CGPA: 8.29

Kolkata, West Bengal

Jalpaiguri Zilla School

01/2015 – 05/2016

Higher Secondary in Science | Percentage: 75.6%

Jalpaiguri, West Bengal

Fanindra Dev Institution

01/2013 – 05/2014

Secondary Education | Percentage: 76.7%

Jalpaiguri, West Bengal

CERTIFICATIONS

- Google Certified Associate Cloud Engineer (ACE)
- Infosys Certified Generative AI Professional – Foundation & Intermediate
- Infosys Certified SQL Developer – Foundation